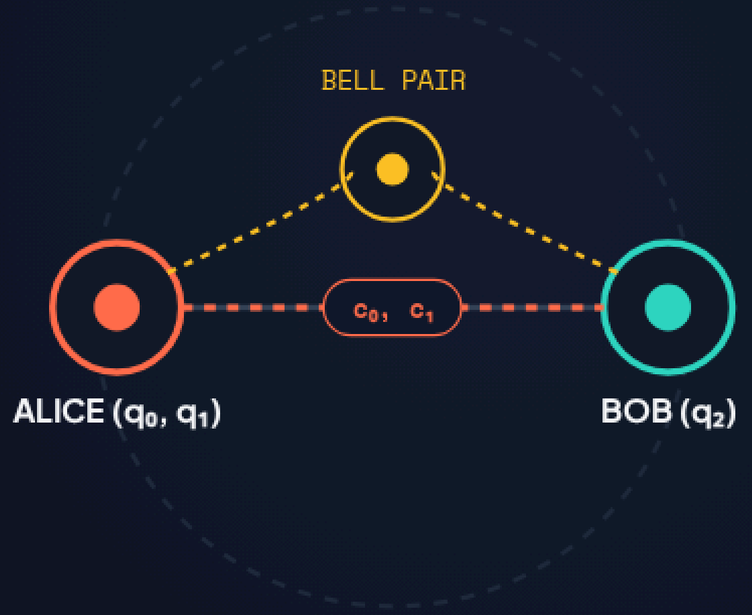
3 QUBITS · 2 CLASSICAL BITS

Quantum *Teleportation* across distance.

Transmitting unknown quantum states using entangled Bell pairs, classical feed-forward, and zero particle movement.

> "Spooky action at a distance." — ALBERT EINSTEIN

Team Members:
Ruben Varghese
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⚠ PHYSICAL CONSTRAINTS

Why quantum states *cannot* be sent classically.

Three fundamental laws of quantum mechanics forbid direct copying or simple classical transmission of arbitrary qubits.

CONSTRAINT 01

The No-Cloning Theorem

It is physically impossible to create an identical copy of an arbitrary unknown quantum state. You cannot duplicate a qubit to send it down two channels.

CONSTRAINT 02

Measurement Wavefunction Collapse

Measuring **α|0⟩ + β|1⟩** yields only a single classical bit (0 or 1). The continuous amplitudes α and β are permanently destroyed upon reading.

CONSTRAINT 03

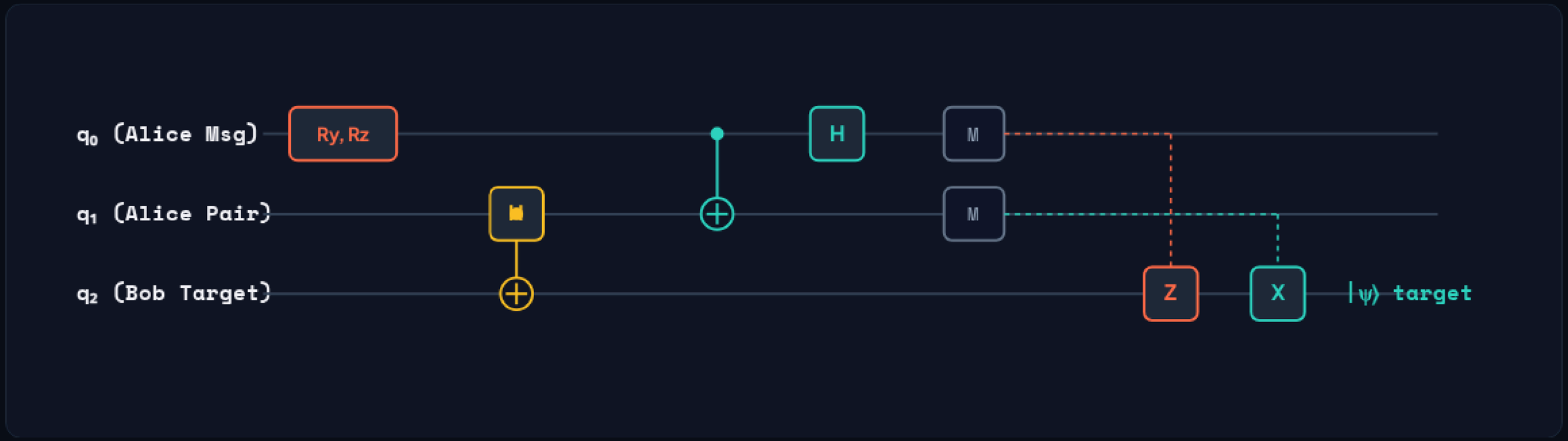
Channel Fragility & Decoherence

Sending actual physical particles (photons) over noisy optical channels causes scattering and photon loss. Once lost, the state cannot be recovered.

CIRCUIT ARCHITECTURE

Five steps, three qubits, *one target state.*

Alice transfers her arbitrary state to Bob by destroying her original qubit and transmitting two classical correction bits.



1. ENCODE
Prepare state on q_0 with Ry, Rz

2. ENTANGLE
Create Bell pair between q_1 & q_2

3. BELL MEASURE
Entangle q_0-q_1 & read 2 bits

4. CORRECT
Bob applies X / Z conditionally

BENCHMARKING VERIFICATION

Testing the six *cardinal states.*

To verify arbitrary state transfer, the protocol was benchmarked across all six poles of the Bloch sphere.

STATE 01 — +Z AXIS

$|0\rangle$

North pole state vector (1, 0)

STATE 02 — -Z AXIS

$|1\rangle$

South pole state vector (0, 1)

STATE 03 — +X AXIS

$|+\rangle$

Equal superposition $(|0\rangle+|1\rangle)/\sqrt{2}$

STATE 04 — -X AXIS

$|-\rangle$

Negative phase $(|0\rangle-|1\rangle)/\sqrt{2}$

STATE 05 — +Y AXIS

$|+i\rangle$

Complex phase $(|0\rangle+i|1\rangle)/\sqrt{2}$

STATE 06 — -Y AXIS

$|-i\rangle$

Negative complex $(|0\rangle-i|1\rangle)/\sqrt{2}$

HARDWARE RESULTS

Fidelity far exceeds *classical guessing.*

Experimental results confirm successful quantum state transfer on both quantum simulators and real IBM hardware.

AVERAGE STATE FIDELITY

94.7%

Successful teleportation across real IBM quantum hardware runs, far exceeding the **~50%** limit achievable by classical guessing.

NOISY SIMULATOR

0.90 – 0.99

High-accuracy fidelity under simulated decoherence and gate error models.

REAL HARDWARE

0.85 – 0.97

Measured on physical IBM backends. Slight degradation caused by T1/T2 relaxation and readout noise.

FUTURE ROADMAP

Honest limits, *next frontiers.*

Entanglement is no longer "spooky". Expanding teleportation from single-qubit circuits to distributed quantum networks.

01

Multi-Qubit States

Extend from single-qubit inputs to multi-qubit registers and entanglement swapping — the core building blocks for quantum repeaters.

02

Inter-Device Networking

Teleport states across distinct physical QPU chips and separate optical backends to move closer to a true distributed quantum internet.

03

Advanced Noise Mitigation

Deploy zero-noise extrapolation and dynamic mid-circuit classical feed-forward to push fidelity closer to theoretical unity.